

MAG — every variable, what it means, and why its weight is what it is

2026-07-27 · 46 features · fitted on 70,681 human decisions from 3,281 clean open-sheet games

Nobody typed these numbers. Each one is fitted: the model is shown a real human decision, all the options that person had, and the weights are nudged until the option they actually clicked comes out most likely. So every weight answers one question — **what makes a human more likely to click this** — and never *what wins*.

ZERO marks a weight whose 95% interval contains zero: the data cannot tell it from no effect.

The last column is the same fit with **popularity removed entirely**, which is the experiment that showed the facts were being crowded out.

What the move does

feature	weight	no-popularity	what it means
eff4	+1.291	+1.193	it hits a 4x weakness
eff2	+0.918	+0.907	it hits a 2x weakness
stab	+0.024 ZERO	+0.247	it matches my own type
bp	-0.030 ZERO	-0.029	it is a powerful move
allyHit	-0.045 ZERO	-0.155	it also hits my own partner, and my partner is not immune to it
accuracy	-0.591	-0.673	how often it hits, on THIS board (snow makes Blizzard certain)
effHalf	-0.863	-0.818	it is resisted
effQuarter	-0.987	-0.986	it is resisted twice over
immune	-2.201	-2.149	it does nothing at all
abilityBlock	-2.287	-2.308	the target probably has an ability that eats it

MOVES THAT COST MORE THAN A TURN

feature	weight	no-popularity	what it means
rechargeTurn	-0.592	-0.632	it costs me the turn AFTER this one
chargeTurn	-1.294	-1.408	it needs a turn to wind up on this board

PIVOTS

feature	weight	no-popularity	what it means
isStatus	+0.487	+0.488	it is a status move
tgtHurt	+0.351	+0.333	the target is already hurt
pivots	-0.584	-0.325	it damages and brings me out
deadWeather	-1.164	-1.384	that weather is already set, so it would fail
deadStall	-1.182	-1.011	I protected last turn, so it would probably fail
deadStatus	-1.400	-1.261	the target already has a status, so it would fail
deadField	-1.902	-1.372	that field effect is already up, so it would fail
pranksterFailsDark	-1.968	-1.966	my Prankster status move is aimed at a Dark type, so it does nothing
deadSide	-2.119	-1.946	that side effect is already up, so it would fail

WHO MOVES FIRST

feature	weight	no-popularity	what it means
defMismatch	+0.370	+0.380	I hit its softer defence
tgtBulk	+0.199	+0.230	the target is bulky
movesFirst	+0.106	+0.108	I move before the target, counting priority, Tailwind and Trick Room
tgtPhysical	-0.264	-0.259	the target attacks physically rather than specially (negative = specially)
priority	-0.409	-0.206	this move cuts the queue (Fake Out, Sucker Punch, Extreme Speed)

DOES IT KILL

feature	weight	no-popularity	what it means
killIsRoll	+0.950	+1.011	it kills some spreads and not others: a roll rather than a read
tgtMayProtect	+0.545	+0.622	how often this target blocks: the biggest reason a sure kill is not one
protectThreatened	+0.266	+0.356	this move protects, and I am facing a kill
koFirst	+0.068	+0.062	it kills, and I move first, so the kill lands before their attack
dmgFrac	-0.071	+0.005	how much of what is left of the target it takes
killsThreat	-0.079	-0.025	it kills the thing that was about to kill me
koTarget	-0.160	-0.150	the odds this really kills it: the worst roll still does, and the move lands

WHAT A MOVE DOES TO THE OTHER PLAYER'S OPTIONS

feature	weight	no-popularity	what it means
volatileOnSelf	+0.222	+0.214	it puts something on me or my side (Substitute, Follow Me, Rage Powder)
volatileOnFoe	-0.404	-0.496	it takes an option away from the target (Taunt, Encore, Disable)

STAT STAGES

feature	weight	no-popularity	what it means
myOffenseStage	+2.004	+1.474	how boosted the stat I am attacking with already is
movesBoostMe	+0.201	+0.242	this move raises one of my own stats
tgtDefenseStage	-0.024 ZERO	-0.019	how boosted the defence I am attacking into already is
movesLowerFoe	-0.370	-1.134	this move lowers one of the target's stats

STATUS THAT ACTUALLY BITES

feature	weight	no-popularity	what it means
statusBites	+0.615	+0.214	the status this inflicts hits a stat the target actually relies on

AM I EVEN GOING TO GET TO DO THIS

feature	weight	no-popularity	what it means
diesBeforeMoving	-0.418	-0.385	I am facing a kill and I do not move first

SWITCHING

feature	weight	no-popularity	what it means
switchSurvives1	+0.233	+0.247	the replacement lives through the hardest thing aimed at me
switchSurvives2	-0.122	-0.045	it lives through that twice
switchFaster	-0.125	-0.148	the replacement outruns the thing that was threatening me
isSwitch	-0.151 ZERO	-2.083	this candidate is a switch, not a move

WHAT THE CROWD DOES

feature	weight	no-popularity	what it means
priorLogP	+0.157	-1.728	it is a popular move

Why the weights look wrong (and mostly are not)

Several weights have a sign that looks absurd — power is negative, accuracy is negative, killing things is negative, Fake Out's priority is negative. Reading any single one as "*MAG dislikes this*" is the mistake. Here is what is actually happening, in each case.

1. Several features measure the same thing, so the fit splits one effect between them

Measured overlap between the damage-ish features:

pair	correlation
bp ↔ dmgFrac	+0.68
dmgFrac ↔ koTarget	+0.59
eff2 ↔ dmgFrac	+0.50
bp ↔ accuracy	+0.43
koTarget ↔ tgtHurt	+0.42

It is the same effect described five ways. `eff4` at **+1.29** takes the credit, and the others are left fitting the leftovers — which go negative. Added together they are right; read one at a time they are nonsense. This is why the model predicts well while its individual numbers look mad.

2. `koTarget` is negative because it is half of a curve

`koTarget` (−0.16) and `killIsRoll` (+0.95) are in the same model and have to be read together. The pull from a kill chance p is $-0.16 \cdot p + 0.95 \cdot 4p(1-p)$, which is a hump:

kill chance	pull
10%	+0.32
40%	+0.84
52%	+0.86 (peak)
90%	+0.19
100%	−0.16

So it was never "*people avoid kills*". **A coin-flip kill is the most-clicked thing in the game.** A certain kill usually means the target was nearly dead and something cheaper was available; a zero-chance kill means the move does not threaten. The big attacks into healthy things sit in the middle.

3. `priority` is negative because `movesFirst` already took the credit

`movesFirst` covers "I act before they do", including `priority`. `priority` therefore only receives the residual — and among moves that go first, the `priority` ones are clicked *less* than the ones that go first through raw Speed. It is not a judgement about Fake Out.

4. The switch weights are noise, and acting on them cost ten points

`switchSurvives1` +0.23, `switchFaster` -0.13, `switchSurvives2` -0.12 with an interval spanning zero. Fitted from a signal the corpus cannot resolve. Enabling switching on those weights was measured across 30,000 paired games:

MAG	opponent	decisive pairs won
cannot switch	cannot switch	81.9%
can switch	can switch	71.6%
can switch	cannot switch	71.6%

The last two being identical is what proves it: the opponent's switching is worth nothing and MAG's own costs ten points. Switching is off by default until a switch policy beats 81.9%.

The reason is visible in the feature list: every switch feature is about **damage and speed**. None of them can see being perish-trapped, Encored, Taunted or confused — the board tracks no volatile statuses at all. So it takes the marginal switches and none of the mandatory ones.

5. `tgtMayProtect` is positive, which is a confound not a preference

+0.55 says people click *into* likely-Protect targets more often. Almost certainly because the species that Protect most are also the common, strong, frequently-targeted ones. The feature is doing real work inside the kill estimate, where it discounts a "guaranteed" kill by the odds the target simply blocks — Protect was **46% of every false kill call** before it was added.

6. `priorLogP` was the largest single weight, and removing it made the model better

"How often people click this move" is the only feature here that is **not a fact about the game**. It is what the previous bot ran on by itself, kept as one term among 46 so its pull could be measured instead of assumed.

Refitting with it removed entirely:

	top-1 accuracy	over the old bot
with popularity	34.6%	+7.4 pts
without popularity	35.2%	+14.3 pts

It predicts human clicks **better without the feature that describes human clicks**, because popularity was crowding out the facts. That is the imitation ceiling showing up as a number.

What the weights cannot tell you

They are fitted to predict clicks, and predicting clicks is not winning. The same 46 features re-optimised for **winning** beat the imitation-fitted version 63.2%. And simply taking the best-scoring move instead of sampling is worth **+9 points** of win rate — five times what a whole night of new features bought.

The features are facts. What they are worth is still fitted to the wrong objective.